

Focused Curation Report — PIK3R4 / VPS15 (Q99570)

Hypothesis (function_assignment / slug protein-phosphorylation-complex-contribution): Human PIK3R4/VPS15 directly contributes to protein phosphorylation (GO:0006468), potentially as a noncatalytic cofactor/scaffold rather than the phosphotransferase.

Source annotation under review: GO:0006468 "protein phosphorylation" (BP), evidence code NAS, reference

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Executive Judgment

Verdict: Weakly supported / over-annotated as written — recommend the GO:0006468 (NAS) annotation be treated as non-core and removed (or, at most, generalized), pending curator verification.

The hypothesis correctly separates two claims that the original annotation conflates:

1. A direct VPS15 protein-kinase molecular function (GO:0004672) — **REFUTED**. Modern cryo-EM of the human class III PI3K complex explicitly designates VPS15 a **pseudokinase** (P 39913640 2025); the seed further notes nucleotide HPLC/MS showing **GTP (not ATP) binding** and **degenerate phosphotransferase geometry**. UniProt Q99570 keeps a legacy "Protein kinase" domain (aa 26–324), an active-site "proton acceptor" (D148) and ATP-binding sites **annotated by similarity**, plus the keywords "Serine/threonine-protein kinase" and "Transferase" — **yet lists no catalytic-activity reaction**, and describes the protein's **FUNCTION** solely as a "Regulatory subunit." The catalytic residues are present as a fold relic, not a working active site.
2. A broad BP "contribution to protein phosphorylation" via the complex — **UNRESOLVED**, and not established for VPS15 specifically by the cited evidence. The protein-substrate phosphorylation in

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(Table I) is a property of the **co-purified GST-p150-VPS34 complex**, with Mn^{2+} preference and **no individual autophosphorylation**. Attribution to VPS15 is not made by the authors; the more parsimonious catalyst is VPS34 (or an associated kinase). In the same paper, p150/VPS15 appears as a **substrate** (Fig 4, in-vivo labeling) — the opposite of being the kinase. The annotation is **NAS**, i.e. author assertion, not a traceable experimental attribution to VPS15. Subsequent direct biochemistry firms this direction: VPS15 is a **ULK substrate**

P 34121209

six ULK-dependent sites, major Ser861) whose phosphorylation regulates VPS34 activity, and a **GTP-binding** conformational regulator of VPS34

P 40537377

— i.e. a phospho-target/regulator, not a protein phosphotransferase.

Most important caveat (from the seed, and respected here): a pseudokinase can still participate noncatalytically. Loss of catalysis $\neq$ loss of all participation. So the correct curation posture is not "VPS15 has nothing to do with phosphorylation," but "the *specific* evidence does not establish a direct VPS15 phosphotransferase function, and a bare BP 'protein phosphorylation' term mis-states VPS15's actual regulatory/scaffold role."

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
P 8999962 (Panaretou 1997, JBC 272:2477)	Direct assay (in vitro), original NAS source	Qualifies / competing	Does p150/VPS15 directly phosphorylate protein substrates?	Affinity-purified complex phosphorylates peptide/protein substrates (Mn ²⁺ -preferring, Table I); no individual autophosphorylation; p150 is a substrate in vivo (Fig 4). Recombinant p150 stimulates VPS34 lipid kinase 2-fold.	Sf9-express GST-p150-human VPS34 complex
P 39913640 (Cook 2025)	Structural (cryo-EM) + biochemistry	Refutes (direct MF)	Is VPS15 a catalytic protein kinase?	VPS15 is a pseudokinase ; its inactive conformation gates VPS34; GTP (not ATP) binding, degenerate active-site geometry (per seed).	Human PI3KC3-C1
P 40537377 (Duttenhefner 2025)	Structural/biochem synthesis	Refutes (direct MF)	Does VPS15 bind GTP and act catalytically?	"the VPS15 pseudokinase domain binds GTP and sequesters its covalently-linked N-terminal myristate"; stabilizes inactive VPS34	Human PI3KC3-C1
P 34121209 (Mercer 2021)	Direct assay (phosphoproteomics + mutants)	Competing / qualifies	Is VPS15 a kinase or a substrate?	VPS15 is a ULK substrate : "six ULK-dependent phosphorylation sites on VPS15, mutation	Mouse embryonic fibroblasts vitro

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
				of which reduces... VPS34 activity in vitro" (major Ser861)	
P 37414850 (Alkhoury 2023)	Mutant/localization	Qualifies (noncatalytic moonlighting)	Does VPS15 have non- complex roles?	Nuclear Vps15 coactivates Bmal1- Clock independently of Vps34 ; promotes Ppat transcription	Mouse liver cells
P 40442316 (Chen 2025)	Structural (cryo-EM)	Supports (scaffold role)	What is VPS15's molecular role?	VPS15 is a scaffold/ interaction hub , contacting FIP200 (ULK1C) and co- assembling the supercomplex; drives ULK1 dimerization.	Human ULK1C:PI3 C1 supercomplex
Catalytic-motif audit (this run, Q99570 vs PKA P17612)	Computational (sequence)	Refutes (direct MF)	Are canonical STK catalytic motifs intact?	Degenerate Gly-rich P-loop; DFG→DFA; only β 3-Lys(K53) and catalytic-Asp(D148) retained	Human sequence a
UniProt Q99570 (database)	Database record	Qualifies	Is a catalytic kinase function annotated?	Kinase domain + active site + ATP- binding by similarity; no catalytic reaction ; FUNCTION = "Regulatory subunit."	Curated human rec
PANTHER PTHR17583 / root PTN000426471 (evolutionary)	Computational/ evolutionary	Qualifies / competing	Should protein kinase activity propagate by ancestry?	Family-root kinase IBA gives evolutionary context but should not be equated with	Ortholog family

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
				retained catalysis in the VPS15 leaf.	

Computed Provenance — VPS15 Kinase-Domain Catalytic-Motif Audit

I retrieved UniProt Q99570 (this run) and audited the kinase domain (aa 26–324) for the canonical Ser/Thr-kinase catalytic elements, benchmarked against canonical PKA (P17612). Executed code + output; table saved as `vps15_catalytic_motif_audit.csv`.

Motif	VPS15 pos	Canonical requirement	VPS15 observed	Verdict
Glycine-rich P-loop (GxGxxG)	~aa30–40	GxGxxG positions ATP phosphates	LGSTRFFKV (no GxGxxG)	DEGENERATE
β3 lysine (VAIK)	K53	Lys pairs α/β phosphates	VVVK (K53 present)	Retained
Catalytic loop (HRD; Asp = proton acceptor)	D148	HRD	HGD (H146,G147,D148)	Asp retained; HRD Arg→Gly
Mg-binding DFG	~D164	DFG	DFA (Gly→Ala)	ALTERED

Benchmark: canonical PKA has an intact P-loop (GTGSFG @51) and DFG @185. **Interpretation (conservative):** two core nucleotide-positioning/catalytic elements (Gly-rich loop; DFG glycine) are degenerate in VPS15, consistent with impaired canonical ATP-dependent phosphotransfer and the pseudokinase designation. The **retained** K53 and D148 are the relics that drive UniProt's "by similarity" ATP-binding/active-site and the PANTHER family-root kinase IBA — i.e. the legacy kinase labels are homology carry-over, not demonstrated catalysis. This sequence audit is corroborative and secondary to the direct cryo-EM structural evidence (

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GO Decision Table (leads — require curator verification)

GO term	Aspect	Current	Evidence verdict	Recommended action
GO:0006468 protein phosphorylation	BP	Present (NAS, P 8999962)	Complex-level activity; VPS15 is substrate not catalyst; VPS15 = pseudokinase	Remove / treat as non-core (or at most generalize)
GO:0004672 / GO:0004674 protein (S/T) kinase activity	MF	Risk via IBA/ISS (PTHR17583 root)	Refuted: degenerate P-loop, DFG→DFA, GTP-binding pseudokinase	Do not assign / remove (evolutionary context only)
GO:0034271/0034272 PI3K complex, class III (type I/II)	CC	—	Strongly supported (structures)	Add/retain
GO:0000045 autophagosome assembly	BP	—	Supported (function, structures)	Retain/add
GO:0016192 vesicle-mediated transport	BP	—	Supported (UniProt function)	Retain/add
Enzyme-activator / kinase-regulator of VPS34 lipid kinase	MF	—	Supported (P 8999962 2-fold; P 34121209)	Consider (more informative than "protein binding")
GTP binding (GO:0005525)	MF	—	Supported by P 39913640 / P 40537377	Consider (curator discretion)

GO Curation Implications (leads — require curator verification)

- **GO:0006468 "protein phosphorylation" (BP), NAS,**

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— **lead: REMOVE or down-rank to non-core.** The cited evidence demonstrates a *complex* activity, not a VPS15 molecular function, and modern structure shows VPS15 is a pseudokinase. A bare BP "protein phosphorylation" on a scaffold subunit mis-attributes a co-subunit's chemistry.

- **GO:0004672 / GO:0004674 "(serine/threonine) protein kinase activity" (MF) — if present via IBA/ISS, lead: REMOVE.** Refuted by pseudokinase evidence; keep only as evolutionary/fold context, not as active MF.
- **Do NOT downgrade to a bare "protein binding" term.** More informative supported terms exist:
- **CC:** GO:0034271/GO:0034272 (phosphatidylinositol 3-kinase complex, class III, type I/II) — supported.
- **BP:** GO:0000045 (autophagosome assembly), GO:0016192 (vesicle-mediated transport) — supported by function/structure.
- **MF:** consider **enzyme regulator / kinase activator** style term for its stimulation of VPS34 lipid kinase

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and note GTP-binding (from

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if curators accept it.

Mechanistic Scope

Immediate molecular function tested: **phosphotransfer onto protein substrates by VPS15 itself.** - **Direct VPS15 activity:** not demonstrated; fold present, catalysis absent (pseudokinase). - **Complex activity (VPS34-containing):** protein-substrate phosphorylation observed in vitro, catalyst unresolved — a *distinct* reaction, not VPS15's MF. - **Downstream/indirect (not the**

tested MF): VPS15 scaffolds PI3KC3, activates VPS34 lipid (not protein) kinase, and is itself a phospho-substrate/regulatory target — regulatory participation in phosphorylation-linked processes, not a phosphotransferase role.

Conflicts and Alternatives

- **Attribution artifact:** complex activity assigned to a subunit that is a substrate in the same paper.
- **Homology carry-over:** UniProt/PANTHER retain kinase/transferase labels by similarity; IBA at the family root risks propagating catalytic MF to a leaf that lost catalysis.
- **Lipid- vs protein-kinase conflation:** the robust, reproducible VPS34-complex chemistry is *lipid* kinase (PtdIns→PtdIns3P); the "protein phosphorylation" claim rests on the weaker Table I assay.
- **Not contamination by default:** an unexpected Mn^{2+} -preferring activity is *not* proof of contamination; it is best logged as unresolved associated activity, per the seed's caution.
- **VPS15 is the phospho-TARGET, not the enzyme:**

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shows VPS15 is a ULK substrate (six ULK-dependent sites; major Ser861), directly realizing the seed's "phosphorylated by another kinase" alternative. Its phosphorylation *regulates* VPS34 lipid-kinase output — a regulatory node in a phosphorylation pathway, not a protein phosphotransferase.

- **Real noncatalytic moonlighting:**

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shows a Vps34-independent nuclear coactivator role (Bmal1-Clock), confirming that "pseudokinase" ≠ "no function," but these functions are not "protein phosphorylation."

Knowledge Gaps

1. **Which subunit is the Table I catalyst?** Checked: abstract + seed description; unresolved. Matters because it decides whether *any* protein-phosphorylation term belongs to VPS15.

Resolve with: recombinant, individually purified catalytically-dead VPS34 vs VPS15 point mutants + substrate assays.

2. Full text of

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nucleotide/geometry data. Checked: abstract (truncated) + seed; full methods not retrieved here. Matters for firmness of "GTP-binding, no phosphotransfer."

3. **Current GO/UniProt evidence codes on the live record** (IBA vs NAS vs ISS). Checked: UniProt JSON (kinase keywords, no catalytic reaction); GO evidence-code audit needed for the exact annotation set.

Discriminating Tests

- **Subunit-resolved kinase assay:** individually purified WT vs kinase-dead (D148A) VPS15 and VPS34 with peptide/protein substrates; $\pm$ Mn^{2+} ; test which subunit carries Table I activity.
- **Nucleotide-loading assay** (HPLC/MS or thermal shift) on isolated human VPS15 pseudokinase domain to confirm GTP-over-ATP preference and absence of γ -phosphate transfer.
- **Structure-guided catalytic-motif audit** (HRD, DFG, VAIK, glycine loop) on Q99570 vs bona fide STKs.
- **Orthology/PANTHER review** to reclassify the IBA MF at the leaf as pseudokinase.

Curation Leads (require curator verification)

- **Candidate action:** Remove/retire GO:0006468 (NAS,

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as a VPS15 direct annotation; do not assign GO:0004672/GO:0004674 MF. Replace informativeness with PI3KC3 CC terms + autophagy/vesicle BP + a VPS34-lipid-kinase *regulator/activator* MF.

- **Candidate reference snippets to verify:**

- P** 39913640

— "The inactive conformation of the VPS15 pseudokinase stabilizes the inactive conformation..."
- P** 40442316

— "...extensive contacts between the FIP200 scaffold subunit of ULK1C and the VPS15, ATG14 and BECN1 subunits of PI3KC3-C1."
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— "Recombinant p150 associated with PtdIns 3-kinase in vitro in a stable manner, resulting in a 2-fold increase in lipid kinase activity."

 - **Suggested question for curators:** Is the NAS GO:0006468 meant to capture VPS15's own MF, or the complex's process participation? If the latter, a co-complex/regulatory annotation is more accurate than "protein phosphorylation."
 - **Suggested experiment:** subunit-resolved catalytically-dead mutant kinase assay (above).

Note: This report deliberately did not re-run the previously completed NVJ analysis; it targets the Panaretou-1997 vs modern-structure phosphotransfer chemistry, as instructed.